

# Automated Mapping of Course Outcomes to Program Outcomes using Natural Language Processing and Machine Learning

Souvik Sengupta

Department of CSE,  
Aliah University, Newtown, Kolkata  
ssg@aliah.ac.in

Asit Kumar Das

Department of CST,  
IIEST, Shibpur, Howrah  
akdas@cs.iests.ac.in

**Abstract**— Curriculum mapping is an essential academic requirement in outcome-based education (OBE) for successfully implementing the academic program. It is a monitoring tool for course coordinators/instructors to align their curricula according to corresponding program outcomes. This mapping is commonly represented by a two-dimensional matrix depicting the relationship between the Course Outcomes (CO) and Program Outcomes (PO), popularly known as CO-PO mapping. This mapping is a challenging task, even for experienced teachers, as it entails knowledge of the standards and practical aspects of effective mapping, which are not well-documented or readily available. Consequently, this paper aims to tackle this challenge by proposing an AI-based automated system for curriculum mapping, which allows teachers to inspect the CO-PO mappings, identify shortcomings, and recommending suggestions for enhancing curriculum mapping. A dataset is prepared by collecting data from publicly available CO-PO mapped curriculums of the Computer Science Engineering program of seven Indian Universities. Since one CO can be mapped to multiple POs, this work employs machine learning and deep learning models with NLP to solve this multi-label classification task. An empirical study is performed to compare the efficiency of different feature extraction and classification methods. The results indicate that the BERT-based model outperforms the other models in the accuracy of mapping.

**Keywords**— Curriculum mapping; Outcome-based Education; CO-PO mapping; NLP; ML; BERT

## I. INTRODUCTION

Mapping course outcomes (CO) to the program outcomes (PO) is an effective method of achieving quality assurance and adherence to accreditation standards to enhance educational outcomes. The POs are set at the program level, considering the students' knowledge and skills. Students should exhibit their academic accomplishments upon successfully completing the courses and the program [1,5,10]. Course Outcomes (COs) refer to the knowledge and skills planned for individual course. These COs, when aligned with the POs, are used to calculate the attainment of each PO, and action plans are developed accordingly at both course and program levels for the academic improvement of the students [6,7,11]. The All India Council for Technical Education (AICTE) has specified 12 program outcomes to ensure that engineering graduates possess the knowledge and skills necessary to meet the demands of the rapidly changing technological landscape. These POs are intended to serve as a guide for engineering educators to design and develop engineering programs that prepare graduates for successful careers. However, Mapping COs to POs is not a trivial task; rather, it can be challenging for several reasons [3]. COs formulated for individual courses

are often specific to that course's subject matter and learning objectives. In contrast, POs represent the overarching knowledge and skills that students should acquire by the end of the program. Thus, identifying the relationship between COs and POs requires a clear understanding of the learning outcomes and objectives at both levels. In addition, due to the absence of any rule-based guidelines, it requires expertise to identify proper techniques for efficient mapping. Another challenge is the volume of data involved in mapping COs to POs. For example, a program may have multiple courses, each with several COs and several POs that students must achieve. Managing and analyzing this data manually can be time-consuming and prone to errors. An automated system for mapping COs to POs can be helpful for accurate analysis of the relationship between COs and POs without human expertise [4]. A trained AI model can identify the appropriate target POs for the COs of any newly developed course. These predictions can help educators to adjust their teaching methods and course design to better align with the program's objectives. In this paper, Machine Learning (ML) and Deep Learning (DL) models are employed in different experimental setups with varying combinations of feature extractions, and a comparative analysis of performances is presented.

## II. RELATED WORKS

Not much research works have been reported on AI-based automatic mapping of course outcomes with program outcomes. There are, however, several works done on automated curricular mapping in general to help the program evaluation. Alshankiti et al. [2] state that without any standard practice and rules for effective mapping of course outcomes with program outcomes, mistakes are common during the mapping, even by experienced teachers. The authors address this challenge by investigating efficacy of mapping rules derived from pre-existing curriculum mappings. It involves inspecting the course mappings, identifying inefficiencies, and providing suggestions for improving the curriculum mapping. The authors proposed a rule-based algorithm for curriculum matrix assessments, where the courses are mapped to PO at three levels: 'introduced', 'reinforcement', and 'emphasized'. They also implemented it with an online application. In a comprehensive study, Watson et al. [8] conducted a thorough evaluation of the existing frameworks through a comparison of the parameters with six medical curriculum maps from Australian educational institutions. They proposed a typology to cover the important choices regarding curricular mapping. The map-typology is employed in this study as a guideline to help curriculum mappers make decisions regarding the development of new maps as well as the enhancement of current ones that have practical

implications for program delivery and evaluation. While most of the proposed curriculum mapping methods are rule-based, only one work, Zaki et al. [9], has tried to map CO-PO using AI and NLP. They advocated that CO-PO mapping ensures quality and proposed an AI-based system that can automate and validate the mapping process. The system was tested on two educational programs and achieved a precision of 83.1% and 88.1%, respectively. The system uses natural language processing to extract keywords from CLOs and PLOs, which are then used for cosine similarity between them. The cosine similarity is used to identify the best mapping between CLOs and PLOs. However, the model suffers from not considering semantic relationships between the words, as no word embedding is performed. In the current work, we introduce a framework for the automatic mapping of CO-PO with the help of NLP and ML. Performances of different combinations of word embedding and ML/DL classifiers are compared and analyzed.

### III. METHODOLOGY

This work employs a simple NLP-based pipeline for the multi-label classification for the task of CO-PO mapping. Fig. 1 depicts the framework of the proposed work. The main components of the framework are data analysis, data preprocessing, feature extraction, and classification. Next, we illustrate the components in detail.

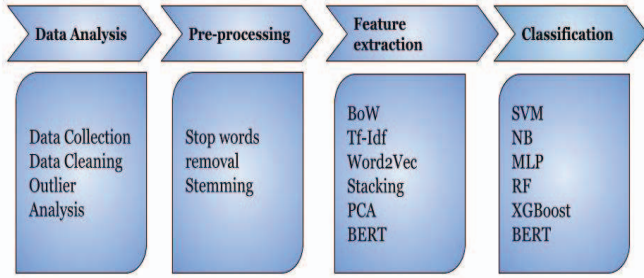


Fig. 1. A framework for automated CO-PO mapping using NLP and ML.

#### A. Data Analysis

It is the initial phase of the pipeline, composed of data collection, data cleaning, and outlier analysis. We have prepared a customized dataset from the collection of publicly available curricula from seven Indian universities. Only AICTE-approved curricula for bachelor-level Computer Science and Engineering (CSE) programs are considered in this study. Although, having a level of correlation, like 1 for low, 2 for medium, and 3 for high, in the mapping matrix is a good indicator for measuring attainment, it is not uniformly available in the data available from different universities. Therefore, we restrain to only binary correlation between CO and PO in this work. A total of ten CSE courses have been selected from the repository based on the coherency of the course outcomes. AICTE has fixed twelve POs for the bachelor level of engineering program. On average, a curriculum contains four to six COs per course, which are mapped with the twelve predefined POs. After the initial data collection, the total number of records in the raw dataset was 264.

Ideally, a well-designed curriculum would have a balanced distribution of course outcomes across the program outcomes, with each course outcome mapping to a subset of the program outcomes that are most directly related and essential for the

course. However, it was observed in the dataset that some COs are mapped to all the POs while some are mapped to very few POs. Although it is not inherently wrong to have such mappings, a more balanced distribution ensures alignment and effectiveness of the curriculum. Mapping COs to all POs indicates a lack of specificity, whereas mapping COs to only a few POs indicates a lack of coherence to the program.

Therefore, outlier analysis is done to improve the quality of the data. The histogram analysis of the total number of mappings by the individual records is shown in Fig. 2. It is observed that 9 records have less than a total of 3 mappings, and 8 records have more than a total of 10 mappings. After discarding these 17 records, the final size of the data set becomes 247. On the other hand, figure 3 shows that the total number of mappings toward a PO is not uniformly distributed. PO1, PO2, PO3, PO4, PO5, and PO12 are more frequently mapped by the COs than the other POs. These six out of the twelve POs have contributed 77% of the total mapping. As automated mapping of CO-PO is a multi-label classification problem with a relatively small data set, this substantial amount of class imbalance needs special attention within the classification model.

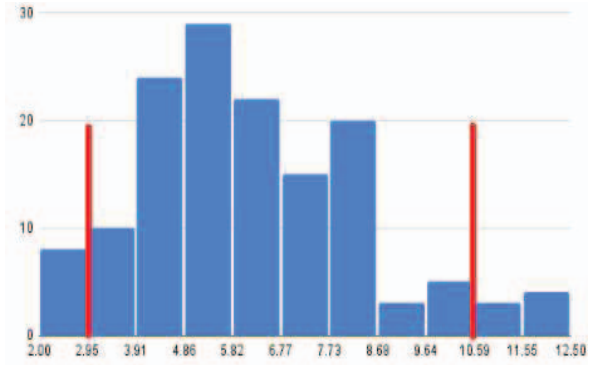


Fig. 2. Histogram analysis of the total number of mappings by Cos.

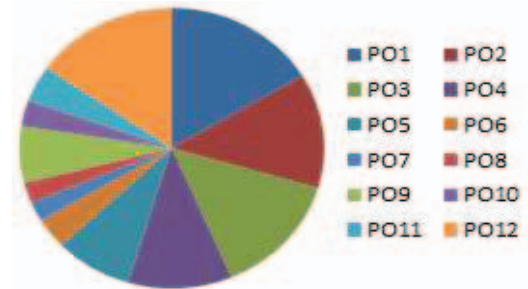


Fig. 3. A Percentage of mapping on POs.

#### B. Preprocessing

The preprocessing phase involves 2 operations- stopword removal and stemming. Stopwords like 'a', 'an', 'is', 'the', 'and', etc., are frequently used words that do not carry any significant meaning. Python 'nltk' library, which contains 153 stopwords from the English dictionary, is used for stopword removal in this work. On the other hand, Stemming involves transforming words into their base or root form by removing suffixes. For instance, the words 'likes', 'likely', and 'liked' are all stemmed from the base form 'like'. The purpose of stemming is to simplify text analysis processes by reducing inflected or derived words to a shared base form to improve

classification accuracy. This work uses 'PoterStemmer' from the Python nltk library to implement this task.

### C. Feature Extraction

Feature extraction is the next important step in the pipeline, which is achieved through word embedding. It represents text data in a numerical or vectorized format, which can then be used as input for machine learning algorithms. Some of the popular feature extraction techniques in NLP are Bag-of-Words (BoW), N-Gram, Term Frequency-Inverse Document Frequency (TF-IDF), Word2Vec, and GloVe. The BoW model represents text as a collection of individual words, disregarding the sequence and word order. TF-IDF is a weighting mechanism that reflects the significance of a word in a specific document as well as in a larger corpus of documents. It calculates a weight for each word by considering its frequency within a document (TF) and inversely scaling it by the number of documents containing the word (IDF). Word2Vec and GloVe are popular word embedding models that also capture semantic relationships between words. They represent words as dense vectors in a continuous space where similar words are closer together. However, these models produce dense vector representations but often generate high-dimensional word embeddings, where each word is represented by a vector of hundreds or thousands of dimensions based on the size of the vocabularies of the training language (approx 20k for English). A popular default value for Word2vec embedding is 300, but it is also high considering the size of our dataset. Principal Component Analysis (PCA) can be used with Word2Vec features for dimensionality reduction by transforming the original features into a lower-dimensional space while retaining as much variance as possible. In this experimental study, we employ three setups with different feature combinations, namely – i) TF-IDF, ii) Word2Vec, and iii) stacked TF-IDF and Word2Vec with PCA.

In addition to conventional word embedding techniques, this work also employs Bidirectional Encoder Representations from Transformers (BERT), a cutting-edge model developed by Google. Unlike the traditional language models, which are trained in a left-to-right or right-to-left manner, BERT trains a deep bidirectional representation by considering both left and right contexts simultaneously, which enables the model to capture more accurate contextual information. The key advantage of BERT lies in its ability to generate contextualized word embeddings. This work shows a performance comparison of three popular extensions of BERT, namely DistillBERT, RoBERTa, and Electra, while transfer learning the pretrained models on our dataset.

### D. Classification Models

Each of the three combinations of features is tested with five ML classifiers – Support Vector Machine (SVM), Naïve Bayes (NB), Multi-Layer Perceptron (MLP), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost). The first three are single classifiers, while the last two are ensemble methods.

*Support Vector Machine (SVM):* SVM algorithm is designed to identify an optimal hyperplane that can effectively separate data points belonging to various classes, while maximizing the margin between them. SVM can effectively address classification challenges, in both linear and non-linear scenarios, by employing suitable kernel functions. SVM is popular for its ability to deal with high-dimensional data and

outliers effectively, and generalize well on small to medium-sized datasets.

*Naïve Bayes (NB):* NB classifier is a simple probabilistic model that applies Bayes' theorem, assuming features' independence. It is a simple yet powerful algorithm that performs well in text classification tasks. NB calculates the probability of data belonging to each class and selects the class with the highest probability. It is especially useful when dealing with large feature spaces and high-dimensional data.

*Multi-Layer Perceptron (MLP):* The MLP is a type of artificial neural network that consists of numerous layers of interconnected nodes. The neural network architecture comprises of an input layer, one or multiple hidden layers, and an output layer. Each neuron in the network applies a non-linear activation function to its weighted inputs, allowing the network to capture complex relationships in the data. MLPs can learn non-linear decision boundaries and are capable of approximating any continuous function given enough training data and appropriate hyperparameter tuning. Training MLPs typically involves forward propagation for prediction and backpropagation for updating the weights based on the error or loss.

*Random Forest (RF):* RF utilizes an ensemble approach by combining numerous decision trees to provide predictions. In this approach, every tree is trained with a random subset of the available training data. Subsequently, while making predictions, the result is obtained by combining the individual predictions made by each tree. RF can overcome the limitations of overfitting and instability and provides improved generalization and robustness to noise.

*eXtreme Gradient Boosting (XGBoost):* The gradient-boosting algorithm is another ensemble model known for its exceptional performance and scalability. XGBoost constructs an ensemble of suboptimal prediction models, commonly decision trees, in a sequential fashion. It optimizes the objective function by iteratively adding models that minimize the overall loss. XGBoost incorporates regularization techniques to prevent overfitting and is efficient in handling large datasets, preventing overfitting, and providing good accuracy.

*Convolutional Neural Network (CNN):* Convolutional Neural Networks exhibit exceptional performance by extracting features from localized input patches, that are salient for sequence processing. 1D convolution layers are translation invariant as a pattern learned at a specific point within a sentence can afterwards be identified at various other positions. Max Pooling and Average Pooling are used for reducing the length of the 1D input by subsampling. This work uses a custom lightweight CNN model with 3 conv1D layers with 2 intermediate maxpooling layers and a final global maxpooling layer. This is followed by 2 dense layers and the classification output. We employ a kernel of size 5 in the convolution blocks, relu loss function at the dense layers, softmax loss function at the output layer, and adam optimizer for the over all model.

## IV. RESULT ANALYSIS

Table I and Table II depict the comparative performances of different machine learning and BERT models, respectively. Since in a multi-label classification task, a model's prediction can be partially correct, we used Hamming loss and F1 score instead of accuracy, precision, and recall as metrics for

measuring performances. Using both Hamming loss and F1 score in a multi-label classification task provides complementary perspectives on the model's performance. Hamming loss offers a holistic evaluation of the label's prediction accuracy, while the F1 score provides label-specific precision and recall information. Together, these metrics provide a comprehensive assessment of the model's performance in a multi-label classification task.

Conventional text features like tf-idf and word2vec are arranged in three experimental settings with the ML models for the classification task. The results indicate that tf-idf appears to be more potent than the word2vec embeddings in our case. This could be due to the fact that the smaller size of the dataset and limited use of the vocabulary (domain-specific) restrict the word2vec model from capturing semantic relationships among the words. On the other hand, the repetition of domain-specific words in individual COs and across the corpus allows the tf-idf model to capture discriminative features based on the word frequencies. It is also evident from Table I that although word2vec independently is not useful for classification, combining tf-idf with word2vec has performed reasonably well. The only exception is the SVM classifier, where the combined feature set yields less accuracy than the tf-idf; in all other cases, the combination gives either greater or at per accuracy with the tf-idf. Among the ML classifiers, there is nothing much to differentiate in terms of performance. However, ensemble methods appear to be more accurate in this multi-label classification task.

On the other hand, Table II shows that BERT-based embeddings with DL models improve the performance by 10% on average than the ML-based models with conventional embeddings. BERT's ability to capture long-range contextual dependencies is preserved in transfer learning, which makes it highly useful in text classification tasks, improving the model's ability to understand and classify text accurately even with our small dataset. From Table 2, it is observed that DistilBERT and RoBERTa performed better than ELECTRA. Understandably, ELECTRA has a smaller model size compared to DistilBERT and RoBERTa and also uses different pre-training methods compared to ELECTRA. DistilBERT is a distilled version of BERT, trained using a combination of unsupervised and supervised learning. RoBERTa is trained with larger batch sizes and more training data compared to other versions of BERT. Among all the experimental setups, Roberta recorded the best performance with an 85.96% F1 score. The primary limitation of this study is that the dataset is domain-specific, and the target classes are predefined. In many disciplines, there is no such list or guideline for program outcomes available. In that case, a novel semi-supervised algorithm may be required, which could be a future extension of this work.

TABLE I. PERFORMANCES OF ML MODELS WITH CONVENTIONAL FEATURES

| Model | Feature     | Hamming Loss | F1 score (Micro Avg.) |
|-------|-------------|--------------|-----------------------|
| SVM   | TF-IDF      | 0.1898       | 0.7764                |
|       | Word2Vec    | 0.2546       | 0.7290                |
|       | Stacked+PCA | 0.2546       | 0.7405                |
| NB    | TF-IDF      | 0.2013       | 0.7651                |
|       | Word2Vec    | 0.4583       | 0.4923                |
|       | Stacked+PCA | 0.1967       | 0.7703                |

|         |             |        |        |
|---------|-------------|--------|--------|
| MLP     | TF-IDF      | 0.2199 | 0.7570 |
|         | Word2Vec    | 0.2384 | 0.7456 |
|         | Stacked+PCA | 0.2199 | 0.7721 |
| RF      | TF-IDF      | 0.1921 | 0.7798 |
|         | Word2Vec    | 0.1990 | 0.7748 |
|         | Stacked+PCA | 0.1944 | 0.7823 |
| XGBoost | TF-IDF      | 0.1967 | 0.7707 |
|         | Word2Vec    | 0.2361 | 0.7384 |
|         | Stacked+PCA | 0.2175 | 0.7733 |

TABLE II. PERFORMANCES OF BERT MODELS

| Model      | Loss   | F1 Score |
|------------|--------|----------|
| DistilBERT | 0.4822 | 0.8333   |
| RoBERTa    | 0.5326 | 0.8596   |
| Electra    | 0.5050 | 0.8067   |

## V. CONCLUSIONS

In this paper, an endeavor is made to automate the process of mapping course outcomes to program outcomes using natural language processing and machine learning. An empirical study is performed on a customized dataset having 247-course outcomes mapped to 12 program outcomes. Different word embedding-based features are used in three combinations with five ML models. It is observed that tf-idf features performed better than the word2vec model when used with ML classifiers. The probable reason could be the smaller size of the vocabulary used by the COs. Sophisticated dense embedding by transformer-like models such as DistilBERT, RoBERTa, and Electra are also explored by means of transfer learning. DistilBERT and RoBERTa performed noticeably better than the conventional feature extraction methods. The highest F1 score of 85.96% has been achieved with the RoBERTa model. In the future, we would like to extend this work to check whether the same pattern is followed in a domain-agnostic environment

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## REFERENCES

- [1] Admuthe, L. S., & Loni, D. Y. (2016). Course Outcome-Program Outcome Mapping Matrix & Attainment-Issues and Model-Based Solutions for Tier II Category. *Journal of Engineering Education Transformations*, 29(Special Issue).
- [2] Alshantqi, A., Alam, T., Benaida, M., Namoun, A., & Taleb, A. (2020). A Rule-based Approach toward Automating the Assessments of Academic Curriculum Mapping. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 11, 12.
- [3] Dargham, J. A., Chekima, A., Omatu, S., & Chin, R. K. Y. (2012, December). Linking program outcomes to the courses outcomes: A top-down approach. In *2012 4th International Congress on Engineering Education* (pp. 1-3). IEEE.
- [4] Phadke, A. S., & Kulkarni, S. S. (2018). Use of network model for analysis of curriculum and its mapping to program outcomes. *Journal of Engineering Education Transformations*, 31(3), 30-34.
- [5] Plaza, C. M., Draugalis, J. R., Slack, M. K., Skrepnek, G. H., & Sauer, K. A. (2007). Curriculum mapping in program assessment and evaluation. *American Journal of Pharmaceutical Education*, 71(2).
- [6] Spencer, D., Riddle, M., & Knewstubb, B. (2012). Curriculum mapping to embed graduate capabilities. *Higher Education Research & Development*, 31(2), 217-231.
- [7] Treadwell, I., Ahlers, O., & Botha, G. C. (2019). Initiating curriculum mapping on the web-based, interactive learning opportunities,

objectives and outcome platform (LOOOP). African Journal of Health Professions Education, 11(1), 27-31.

- [8] Watson, E. G. S., Steketee, C., Mansfield, K., Moore, M., Dalziel, B., Damodaran, A., ... & Hu, W. (2020). Curriculum mapping for health professions education: A typology. Focus on Health Professional Education: A Multi-Professional Journal, 21(1), 91-113.
- [9] Zaki, N., Turaev, S., Shuaib, K., Krishnan, A., & Mohamed, E. A. (2022). Automating the Mapping of Course Learning Outcomes to

Program Learning Outcomes using Natural Language Processing for Accurate Educational Program Evaluation.

- [10] Sengupta, S. (2023) Towards Finding a Minimal Set of Features for Predicting Students' Performance Using Educational Data Mining. IJ. Modern Education and Computer Science, 3(1), 44-54.
- [11] Sengupta, S. (2022). Possibilities and challenges of online education in India during the COVID-19 pandemic. International Journal of Web-Based Learning and Teaching Technologies (IJWLTT), 17(4), 1-11.